

Lecture 6: Inference and Interpretation

25117 - Econometrics

Universitat Pompeu Fabra

October 23rd, 2024

What we learned in the last lesson

- Omitted variable bias occurs when an omitted variable (a) is correlated with an included regressor and (b) is a determinant of Y .
- The multiple regression model is a linear regression model that includes multiple regressors, $X_1, \dots, X_k$. Associated with each regressor is a regression coefficient, $\beta_1, \dots, \beta_k$.
- The coefficient β_1 is the expected difference in Y associated with a one-unit difference in X_1 , holding the other regressors constant. The other regression coefficients have an analogous interpretation.

What we learned in the last lesson

- The coefficients in multiple regression can be estimated by OLS. When the four least squares assumptions are satisfied, the OLS estimators of the causal effect are unbiased, consistent, and normally distributed in large samples.
- The role of control variables is to hold constant omitted factors so that the variable of interest is no longer correlated with the error term. Properly chosen control variables can eliminate omitted variable bias in the OLS estimate of the causal effect of interest.
- Perfect multicollinearity, which occurs when one regressor is an exact linear function of the other regressors, usually arises from a mistake in choosing which regressors to include in a multiple regression. Solving perfect multicollinearity requires changing the set of regressors.

Interpreting Single Coefficients in Multiple Regression Analysis

- Hypothesis tests and confidence intervals for a single coefficient in multiple regression follow the same logic and recipe as for the slope coefficient in a single-regressor model.
- $\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{var}(\hat{\beta}_1)}} \sim \mathcal{N}(0, 1)$ by the Central Limit Theorem
- Thus hypotheses on β_1 can be tested using the usual t-statistic, and confidence intervals are constructed as $\hat{\beta}_1 \pm 1.96 \times SE(\hat{\beta}_1)$
- So too for $\beta_1, \beta_2, \dots, \beta_k \dots$

Interpreting Single Coefficients in Multiple Regression Analysis

```
. reg testscore frpm_frac_s ed_frac_bachplus, robust
```

Linear regression

Number of obs = 500

F(2, 497) = 241.13

Prob > F = 0.0000

R-squared = 0.5221

Root MSE = 41.772

testscore	Robust		t	P> t	[95% conf. interval]	
	Coefficient	std. err.				
frpm_frac_s	-122.532	9.458738	-12.95	0.000	-141.116	-103.9479
ed_frac_bachplus	89.04404	19.42138	4.58	0.000	50.88591	127.2022
_cons	806.3163	9.468789	85.16	0.000	787.7125	824.9201

Tests of Joint Hypotheses

Imagine we are interested in estimating

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{frpmfracs}_i + \beta_2 \text{edfracbachplus}_i + \beta_3 \text{strs}_i + u_i$$

Where strs_i is the student-teacher ratio in school i . Your theory tells you that, together with the share of subsidized meals, it captures *school resources*.

You want to test whether *school resources* matter for test scores.

$$\begin{array}{ll} H_0 & := \beta_1 = 0 \text{ AND } \beta_3 = 0 \\ \text{VS } H_1 & := \beta_1 \neq 0 \text{ OR } \beta_3 \neq 0 \text{ OR both} \end{array}$$

Tests of Joint Hypotheses

$$\begin{aligned} H_0 &:= \beta_1 = 0 \text{ AND } \beta_3 = 0 \\ \text{VS } H_1 &:= \beta_1 \neq 0 \text{ OR } \beta_3 \neq 0 \text{ OR both} \end{aligned}$$

- A joint hypothesis specifies a value for two or more coefficients, that is, it imposes a restriction on two or more coefficients.
- In general, a joint hypothesis will involve q restrictions. In the example, $q = 2$, and the two restrictions are $\beta_1 \neq 0$ and $\beta_3 \neq 0$
- A “common sense” idea is to reject if either of the individual t-statistics exceeds 1.96 in absolute value.
- But this “one at a time” test isn’t valid: the resulting test rejects too often under the null hypothesis (more than 5%) !

Why can't we just test the coefficients one at a time?

- Because the rejection rate under the null isn't 5%.
- We'll calculate the probability of incorrectly rejecting the null using the “common sense” test based on the two individual t-statistics.
- To simplify suppose that X_1 and X_3 are independently distributed. Let t_1 and t_3 be

$$t_1 = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \quad \& \quad t_3 = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)}$$

- The “one at time” test would say we reject $H_0 := \beta_1 = \beta_3 = 0$ if:

$$|t_1| > 1.96 \quad \text{AND/OR} \quad |t_3| > 1.96$$

- What is the probability that this “one at a time” test rejects H_0 when H_0 is actually true (i.e., false positive)?

Why can't we just test the coefficients one at a time?

The probability of incorrectly rejecting the null hypothesis using the “one at a time” test (we want it to be 5% and assume $t_1 \perp t_3$):

$$\begin{aligned} & Pr_{H_0}[|t_1| > 1.96 \text{ OR } |t_3| > 1.96] \\ = & 1 - Pr_{H_0}[|t_1| \leq 1.96 \text{ AND } |t_3| \leq 1.96] \\ = & 1 - Pr_{H_0}[|t_1| \leq 1.96 \times |t_3| \leq 1.96] \end{aligned}$$

since t_1 and t_3 are independent by assumption.

$$\begin{aligned} = & 1 - Pr_{H_0}[|t_1| \leq 1.96] \times Pr_{H_0}[|t_3| \leq 1.96] \\ = & 1 - (.95)^2 \\ = & 0.0975 \end{aligned}$$

Which is way larger than the 5%-critical value associated with the single t-test!
Note that we have assumed independence but in fact, its size of the test depends on the correlation between t_1 and t_3 , i.e., between $\hat{\beta}_1$ and $\hat{\beta}_3$.

The F-statistic

- **The solution:** use a different test statistic designed to test both β_1 and β_3 at once: the F-statistic
- The F-statistic tests all parts of a joint hypothesis at once.
- Formula for the special case of the joint hypothesis $\beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$ in a regression with two restrictions:

$$F = \frac{1}{2} \frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2}$$

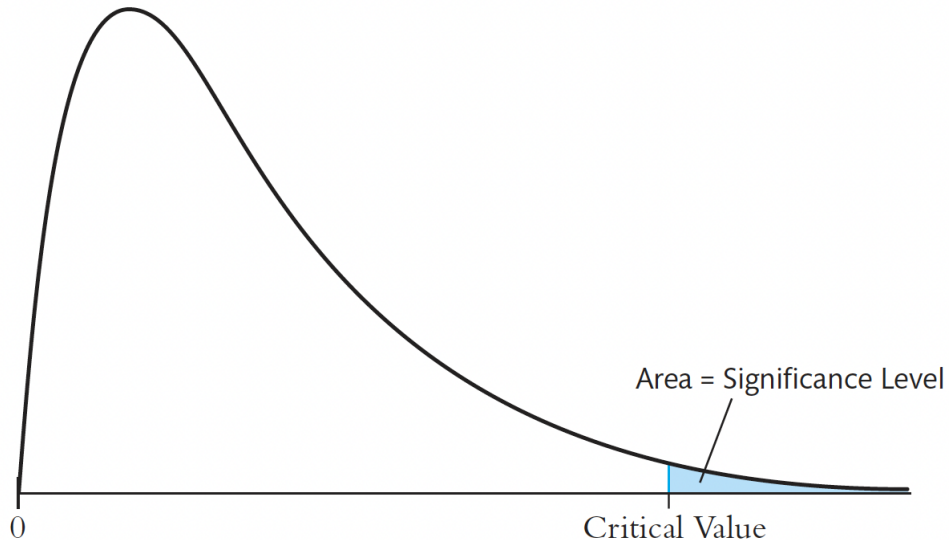
- With t_1 and t_2 being independently distributed, the large-sample distribution of the F-statistic is the distribution of the average of two independently distributed squared standard normal random variables...
- With q restrictions, the formula is more convoluted and requires softwares to be calculated
- We reject the null when F is large. But how large? What is the distribution of the F-stat?

The $F_{q,n-k-1}$ distribution

- The F distribution is tabulated in many places
- As $n \rightarrow \infty$, $F_{q,n-k-1}$ distribution asymptotes to the χ_q^2/q distribution
- $F_{q,\infty}$ is the same as the χ_q^2/q
- The chi-squared distribution with q degrees of freedom (χ_q^2) is defined to be the distribution of the sum of q independent squared standard normal random variables.
- For q not too big and $n \geq 100$, the $F_{q,n-k-1}$ and the χ_q^2/q are essentially identical
- Many regression packages (including STATA) compute p-values of F-statistics using the F-distribution
- You will encounter the F distribution in published empirical work. Here, $F_{q,\infty}$:

Degrees of Freedom	10%	5%	1%
1	2.71	3.84	6.63
2	2.30	3.00	4.61
3	2.08	2.60	3.78
4	1.94	2.37	3.32
5	1.85	2.21	3.02

The $F_{q,\infty}$ distribution



Interpreting Single Coefficients in Multiple Regression Analysis

```
. reg testscore frpm_frac_s ed_frac_bachplus str_s, robust
```

Linear regression

Number of obs = 500
F(3, 496) = 155.10
Prob > F = 0.0000
R-squared = 0.4775
Root MSE = 42.956

testscore	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
frpm_frac_s	-128.6853	10.81701	-11.90	0.000	-149.9382	-107.4325
ed_frac_bachplus	60.72873	20.42014	2.97	0.003	20.60808	100.8494
str_s	.5643052	.5763171	0.98	0.328	-.5680185	1.696629
_cons	802.045	19.84183	40.42	0.000	763.0606	841.0294

```
test str_s frpm_frac_s
```

```
( 1) str_s = 0  
( 2) frpm_frac_s = 0
```

```
F( 2, 496) = 78.12  
Prob > F = 0.0000
```

We can reject the null hypothesis that *neither* the share of subsidized meals *nor* the student-teacher ratio have an effect on test scores.

The Homoskedastic-only F-statistic

There is a simple formula for the F-statistic that holds only under homoscedasticity (so it isn't very useful) but which nevertheless might help you understand what the F-statistic is doing.

When the errors are homoscedastic, there is a simple formula for computing the “homoscedasticity-only” F-statistic:

- 1 Run two regressions, one under the null hypothesis (the “restricted” regression) and one under the alternative hypothesis (the “unrestricted” regression).
- 2 Compare the fits of the regressions — the R^2 's — if the “unrestricted” model fits sufficiently better, reject the null

Example: Are the coefficients on *strs* and *frpmfracs* zero?

Restricted population regression (that is, under H_0):

$$TestScore_i = \beta_0 + \beta_2 edfracbachplus_i + u_i$$

Unrestricted population regression (that is, under H_1):

$$TestScore_i = \beta_0 + \beta_1 frpmfracs_i + \beta_2 edfracbachplus_i + \beta_3 strs_i + u_i$$

The fit will be better (higher R^2) in the unrestricted regression. But by how much should it be higher for *strs* and *frpmfracs* to be deemed statistically significantly different from zero?

The Homoskedastic-only F-statistic

$$F_q^{\text{Homoskedastic-only}} = \frac{(R_{unrestricted}^2 - R_{restricted}^2)/q}{(1 - R_{unrestricted}^2)/(n - k_{unrestricted} - 1)}$$

where

- $R_{unrestricted}^2$ is the R^2 for the unrestricted regression
- $R_{restricted}^2$ is the R^2 for the restricted regression
- q is the number of restrictions under H_0
- $k_{unrestricted}$ is the number of regressors in the unrestricted regression

So the bigger the difference between the restricted and unrestricted R^2 – i.e., the greater the improvement in fit by adding the variables in question – the larger the homoskedasticity-only F.

The Homoskedastic-only F-statistic

Restricted population regression (that is, under H_0):

$$TestScore_i = 705.157 + 203.582 \times edfracbachplus_i + u_i$$

Standard errors: (5.106) (21.472)

with $R^2_{restricted} = 0.2320$

Unrestricted population regression (that is, under H_0):

$$TestScore_i = 802.045 - 128.685 \times frpmfracs_i + 60.729 \times edfracbachplus_i + 0.564 \times strs_i + u_i$$

Standard errors: (19.841) (10.817) (20.420) (0.576)

with $R^2_{unrestricted} = 0.4775$

The Homoskedastic-only F-statistic

$$F_q^{Homoskedastic-only} = \frac{(0.4775 - 0.2320)/2}{(1 - 0.4775)/(500 - 3 - 1)} = 116.5244$$

whereas the heteroskedasticity-robust F is

$$F_q = 78.12$$

- If the errors are a) i.i.d., b) homoskedastic, and c) normally distributed, then the homoskedasticity-only F-statistic follows an $F_{q,n-k-1}$ distribution
- But if the errors are heteroscedastic, the large-sample distribution of the homoscedasticity-only F-statistic is **not** χ_q^2/q

Why the Homoskedastic-only F-statistic?

- The theory of the homoskedasticity-only F-statistic and the $F_{q,n-k-1}$ distributions rests on implausibly strong assumptions
- These statistics date to the early 20th century when data sets were small and computers were people...
- The F-statistic and $F_{q,n-k-1}$ distribution were major breakthroughs: an easily computed formula; a single set of tables that could be published once, then applied in many settings; and a precise, mathematically elegant justification.
- The strong assumptions were a minor price for this breakthrough.
- But with modern computers and large samples we can use the heteroskedasticity-robust F-statistic and the $F_{q,\infty}$ distribution, which only require the four least squares assumptions
- This historical legacy persists in modern software, in which homoskedasticity-only standard errors (and F-statistics) are the default, and in which p-values are computed using the $F_{q,\infty}$ distribution

Summary

- The homoskedasticity-only F-statistic and the $F_{q,n-k-1}$ distributions are justified only under very strong conditions – stronger than are realistic in practice.
- You should use the heteroskedasticity-robust F-statistic, with $F_{q,\infty}$ (i.e., χ_q^2/q) critical values
- For $n \geq 100$, the F-statistic distribution is essentially the χ_q^2/q distribution
- The homoskedasticity-only F-statistic is important historically (and thus in practice), and can help intuition, but isn't valid when there is heteroskedasticity

Testing Single Restrictions on Multiple Coefficients

Consider the following specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

and the null and alternative hypothesis,

$$H_0 := \beta_1 = \beta_2 \quad \text{VS} \quad H_1 := \beta_1 \neq \beta_2$$

This null imposes a single restriction on multiple coefficients – it is not a joint hypothesis with multiple restrictions!

Here are two methods for testing single restrictions on multiple coefficients:

- 1 Rearrange the regressors so that the restriction becomes a restriction on a single coefficient in an equivalent regression; or,
- 2 Perform the test directly (Some software, including STATA, lets you test restrictions using multiple coefficients directly)

Rearranging

From the following specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Add and subtract $\beta_2 X_{1i}$:

$$\begin{aligned} Y_i &= \beta_0 + (\beta_1 - \beta_2) X_{1i} + \beta_2 (X_{2i} + X_{1i}) + u_i \\ &= \beta_0 + \gamma_1 X_{1i} + \beta_2 W_i + u_i \end{aligned}$$

Then test:

$$H_0 := \gamma_1 = 0 \text{ VS } H_1 := \gamma_1 \neq 0$$

The original and rearranged regressions have the same R^2 , the same predicted values, and the same residuals, but the problem is now a simpler one.

Direct Testing (here, with STATA)

```
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```

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_cons	802.045	19.84183	40.42	0.000	763.0606	841.0294

```
. test frpm_frac_s = ed_frac_bachplus
```

```
( 1) frpm_frac_s - ed_frac_bachplus = 0
```

F(1, 496) = 176.35
Prob > F = 0.0000

Confidence Sets for Multiple Coefficients

Consider the following regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

What is a joint confidence set for β_1 and β_2 ?

A 95% joint confidence set is:

- A set-valued function of the data that contains the true coefficient(s) in 95% hypothetical repeated samples.
- Equivalently, the set of coefficient values that cannot be rejected at the 5% significance level

You can find a 95% confidence set as the set of (β_1, β_2) that cannot be rejected at the 5% level using an F-test (why not just combine the two 95% confidence intervals?).

Confidence Sets for Multiple Coefficients

- Let $F(\beta_{1,0}, \beta_{2,0})$ be the (heteroskedasticity-robust) F-statistic testing the hypothesis that $\beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$
- 95% confidence set is defined by $\{\beta_{1,0}, \beta_{2,0} : F(\beta_{1,0}, \beta_{2,0}) < 3.00\}$
- ... where 3.00 is the 5% critical value of the $F_{2,\infty}$ distribution
- This set has a coverage rate of 95% because the test on which it is based (the test it “inverts”) has a size of 5%
- ... i.e., 5% of the time, the test incorrectly rejects the null when the null is true, so 95% of the time it does not; therefore the confidence set constructed as the nonrejected values contains the true value 95% of the time (in 95% of all samples).

Confidence Sets for Multiple Coefficients

$$\begin{aligned} & \left\{ \beta_{1,0}, \beta_{2,0} : F(\beta_{1,0}, \beta_{2,0}) < 3.00 \right\} \\ \Leftrightarrow & \left\{ \beta_{1,0}, \beta_{2,0} : F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right) < 3.00 \right\} \end{aligned}$$

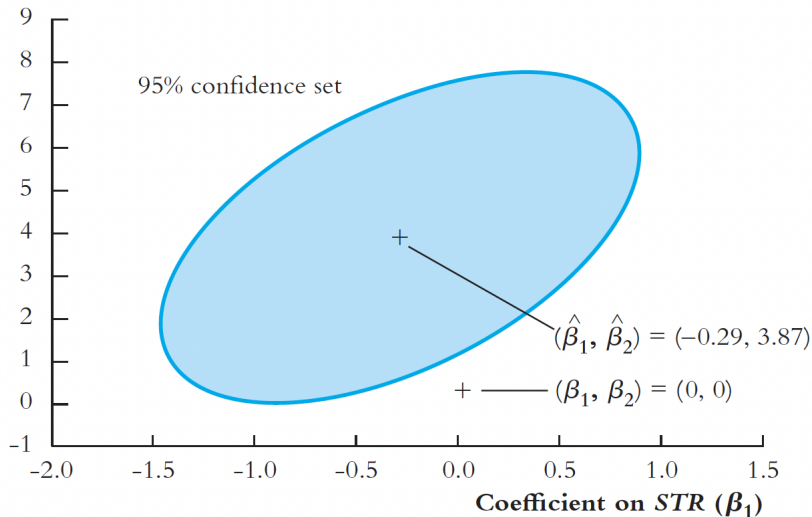
That is,

$$F = \frac{1}{2(1 - \hat{\rho}_{t_1, t_2}^2)} \times \left[\left(\frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \right)^2 + \left(\frac{\hat{\beta}_2 - \beta_{2,0}}{SE(\hat{\beta}_2)} \right)^2 + 2\hat{\rho}_{t_1, t_2}^2 \left(\frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \right) \left(\frac{\hat{\beta}_2 - \beta_{2,0}}{SE(\hat{\beta}_2)} \right) \right]$$

thus the boundary of the set defined by $F < 3.00$ is an ellipse.

Confidence Sets for Multiple Coefficients

Coefficient on *Expn* (β_2)



How to decide what variables to include

- ① Identify the variable of interest
- ② Think of the omitted causal effects that could result in omitted variable bias
- ③ Include those omitted causal effects if you can or, if you can't, include variables correlated with them that serve as control variables.
 - The control variables are effective if the conditional mean independence assumption plausibly holds, that is, if u is uncorrelated with the share of subsidized meals once the control variables are included.
 - This results in a “base” or “benchmark” model.
- ④ Also specify a range of plausible alternative models, which include additional candidate variables.
- ⑤ Estimate your base model and plausible alternative specifications (“sensitivity checks”).
 - Does a candidate variable change the coefficient of interest β_1 ?
 - Is a candidate variable statistically significant?
 - Use judgment, not a mechanical recipe...
 - Don't just try to maximize the R^2 !

What the R^2 tells you, and what it doesn't

It is easy to fall into the trap of maximizing the R^2 and $\bar{R}^2$ but this loses sight of our real objective: an unbiased estimator of the class size effect.

- A high R^2 (or $\bar{R}^2$) means that the regressors explain the variation in your data
- A high R^2 (or $\bar{R}^2$) **does not** mean that you have eliminated omitted variable bias
- A high R^2 (or $\bar{R}^2$) **does not** mean that you have an unbiased estimator of a causal effect β_1
- A high R^2 (or $\bar{R}^2$) **does not** mean that the included variables are statistically significant — this must be determined using hypotheses tests.

Summary

- Multiple regression allows you to estimate the effect on Y of a change in X_1 , holding other included variables constant.
- If you can measure a variable, you can avoid omitted variable bias from that variable by including it.
- If you can't measure the omitted variable, you still might be able to control for its effect by including a control variable.
- There is no simple recipe for deciding which variables belong in a regression – you must exercise judgment.
- One approach is to specify a base model – relying on a priori reasoning – then explore the sensitivity of the key estimate(s) in alternative specifications.

Material I

- **Textbooks:**
 - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 7